南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: College Physics II /A **Dept.: Physics**

Exam Duration: 2 hours **Exam Paper Setter: Physics Teaching Team**

Question No.	1	2	3	4	5	6	7	8	9	10
Score	30	6	10	10	12	10	12	10		

This exam paper contains _8_____questions and the score is _100______ in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Q1. Multiple Choice Questions (3 points each), there is only one correct answer for each question.

1) A 5.0 C charge is 10 m from a –2.0 C charge. The electrostatic force is on the positive charge is:
A) 9.0×10^8 N toward the negative charge
B) 9.0×10^8 N away from the negative charge
C) 9.0×10^9 N toward the negative charge
D) 9.0×10^9 N away from the negative charge
E) none of these

2) Two point particles, one with charge $+8 \times 10^{-9}$ C and the other with charge -2×10^{-9} C, are separated by 4 m. The magnitude of electric field midway between them is:
A) 9×10^9 N/C B) 13,500 N/C C) 135,000 N/C
D) 6×10^{-9} N/C E) 22.5 N/C

3) A battery of emf 24 V is connected to a 6.0 Ω resistor. As a result, current of 3.0 A exists in the resistor. The rate at which energy is being dissipated(损耗) in the battery is:
A) 3.0 W B) 6.0 W C) 18 W D) 54 W E) 72 W

4) A particle with charge 5.5×10^{-6} C is 3.5 cm from a particle with charge -2.3×10^{-8} C. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:
A) 3.3×10^{-2} J B) -3.3×10^{-2} J C) 9.3×10^{-1} J
D) -9.3×10^{-1} J E) 0 J

5) The capacitance of a parallel-plate capacitor can be increased by:
A) increasing the charge B) decreasing the charge
C) increasing the plate separation D) decreasing the plate separation
E) decreasing the plate area

6) A wire has an electric field of 6.2 V/m and carries a current density of 2.4×10^8 A/m^2. What is its resistivity ρ?

A) 6.7×10^{-10} Ω·m B) 1.5×10^{-8} Ω·m C) 2.6×10^{-8} Ω·m
D) 3.9×10^7 Ω·m E) 1.5×10^9 Ω·m

7) The magnetic torque exerted on a flat current-carrying loop of wire (magnetic dipole) by a uniform magnetic field $\vec{B}$ is:

A) maximum when the plane of the loop is perpendicular to $\vec{B}$
B) maximum when the plane of the loop is parallel to $\vec{B}$
C) dependent on the shape of the loop for a fixed loop area
D) independent of the orientation of the loop
E) such as to rotate the loop around the magnetic field lines

8) A toroid with a square cross section carries current i. The magnetic field has its largest magnitude:

A) at the center of the hole
B) just inside the toroid at its outer surface
C) just inside the toroid at its inner surface
D) at any point inside (the field is uniform)
E) at none of the above

9) A conducting rectangular solid of dimensions dx = 5.00 m, dy = 3.00 m, and dz = 2.00 m moves with a constant velocity $\vec{v} = (20.0\text{m/s})\hat{i}$ in a uniform magnetic field $\vec{B} = (30.0\text{mT})\hat{j}$. The resulting potential difference across the solid is:

A) 1.20 V B) 1.80 V C) 3.00 V D) 0 E) 6.00 V

10) Fig.1 shows, in cross section, three infinitely large nonconducting sheets on which charge is uniformly spread. The surface charge densities are $\sigma_1 = +2.00$ μC/m^2, $\sigma_2 = -4.00$ μC/m^2, and $\sigma_3 = +5.00$ μC/m^2, and distance L = 1.50 cm. The net electric field at point P is:

A) $(-5.65 \times 10^5$ N/C$)\hat{j}$
B) $(5.65 \times 10^5$ N/C$)\hat{j}$
C) $(3.95 \times 10^5$ N/C$)\hat{j}$
D) $(-3.95 \times 10^5$ N/C$)\hat{j}$
E) $(-7.90 \times 10^5$ N/C$)\hat{j}$

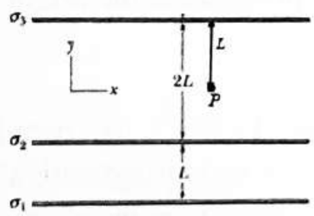

Fig. 1

Q2. Blank-filling Questions. (Complete each of the following questions according to your calculation)

1. (3 points) A proton is a distance $d/2$ directly above the center of a square of side d. The magnitude of the electric flux through the square is______.

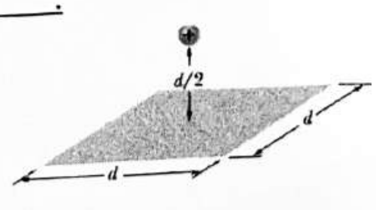

Fig. 2-1

2. (3 points) The magnetic field at any point in the xy plane is given by $\vec{B} = Ar^2\hat{r}\times\hat{k}$, where $\hat{r}$ is the unit vector along the direction of position vector of the point, A is a constant, and $\hat{k}$ is a unit vector in the $+z$ direction. The net current through a circle of radius R, in the xy plane and centered at the origin as shown in Fig.2-2 is given by______. (in terms of A, R and μ_0)

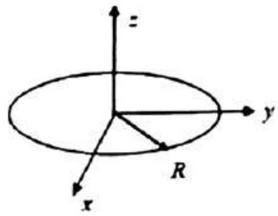

Fig. 2-2

Text Questions: (Please show the solving process in detail)

Q3. (10 points) In Fig.3, a nonconducting rod of length L= 8.15 cm has a charge q = –4.23 fC uniformly distributed along its length.

a) What is the linear charge density of the rod?

b) What are the magnitude and direction (relative to the positive direction of the x axis) of the electric field produced at point P, at distance a = 6.00 cm from the rod?

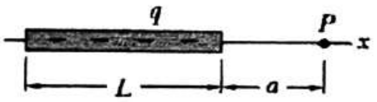

Fig. 3

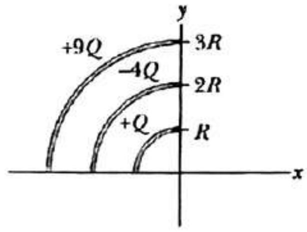

Fig. 4

Q4. (10 points) Fig.4 shows three circular arcs centered on the origin of a coordinate system. On each arc, the uniformly distributed charge is given in terms of Q = 2.00 mC. The radii are given in terms of R = 10.0 cm. With V = 0 at infinity, what is the net electrical potential at the origin due to the arcs?

Q5. (12 points) In Fig.5, two parallel-plate capacitors are connected in parallel across a 10 V battery. One is filled with air, and the other is filled with a dielectric for which κ = 3.00; both capacitors have a plate area of 5.00×10^{-3} m^2 and a plate separation of 2.00 mm. (Neglect the fringing of the electric field at the edges of the plates)

a) How much charge is stored on the parallel-plate capacitor C_1 and C_2, respectively?

b) Find the magnitude of the electric field within the dielectric of capacitor C_1.

c) What is the induced charge on the top surface of the dielectric?

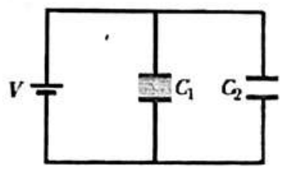

Fig. 5

Q6. (10 points) In Fig.6, two infinitely long wires carry equal current i. Each follows a 90° arc on the circumference (圆周) of the same circle of radius R. In terms of i, R and μ_0, find the magnitude of magnetic field at the center of the circle.

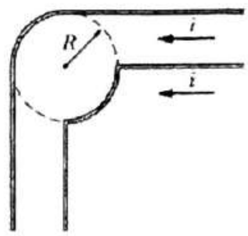

Fig. 6

Q7. (12 points) In Fig.7, a conducting solid sphere of radius a = 2.00 cm is concentric with a spherical nonconducting shell of inner radius $b = 2.00a$ and outer radius $c = 2.40a$. The sphere has a net charge q_1 = 45.0 fC; the shell has a positive volume charge density $\rho = A/r$, where A is a constant and r is the distance from the center of the shell.

a) What is the magnitude of the electric field at radial distance $r = a/2$?

b) What value should A have if the magnitude of electric field in the shell is to be same?

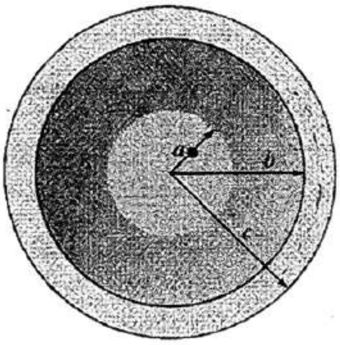

Fig. 7

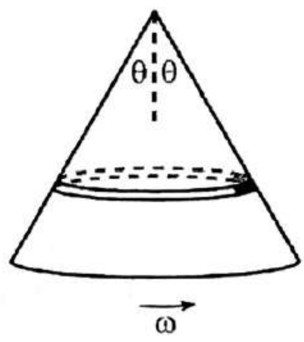

Fig. 8

Q8. (10 points)

a) An infinite cylindrical shell with radius R and surface charge density σ rotates around its symmetry axis with angular frequency ω. Find the magnetic field inside the cylinder.

b) In Fig.8, a hollow cone (like a party hat) has vertex angle 2θ, slant height L, and surface charge density σ. It rotates around its symmetry axis with angular frequency ω. What is the magnetic field at the tip?(*Hint*: A point P located a distance z along the circular coil's perpendicular central axis is $\vec{B}(z) = \frac{\mu_0}{2\pi}\frac{\vec{\mu}}{z^3}$)

Q1

AECBD, CBCAD

Q2

1. $e/6\varepsilon_0 = 3\times10^{-9}$ V.m **3 points**

2. $2\pi AR^3/\mu_0$ **3 points**

Q3 **10 points**

(a) The linear charge density is the charge per unit length of rod. Since the charge is uniformly distributed on the rod,

$$\lambda = \frac{-q}{L} = \frac{-4.23\times10^{-15}\ \text{C}}{0.0815\ \text{m}} = -5.19\times10^{-14}\ \text{C/m}.$$

(b) We position the x axis along the rod with the origin at the left end of the rod, as shown in the diagram.

Let dx be an infinitesimal length of rod at x. The charge in this segment is $dq = \lambda\, dx$.

The charge dq may be considered to be a point charge. The electric field it produces at point P has only an x component, and this component is given by

$$dE_x = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dx}{(L+a-x)^2}$$

The total electric field produced at P by the whole rod is the integral

$$E_x = \frac{\lambda}{4\pi\varepsilon_0}\int_0^L \frac{dx}{(L+a-x)^2} = \frac{\lambda}{4\pi\varepsilon_0}\left.\frac{1}{L+a-x}\right|_0^L = \frac{\lambda}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{L+a}\right)$$
$$= \frac{\lambda}{4\pi\varepsilon_0}\frac{L}{a(L+a)} = -\frac{1}{4\pi\varepsilon_0}\frac{q}{a(L+a)},$$

upon substituting $-q = \lambda L$. With $q = 4.23 \times 10^{15}$ C, L =0.0815 m and a = 0.060 m, we obtain $E_x = -4.48\times10^{-3}$ N/C , or $|E_x| = 4.48\times10^{-3}$ N/C . The negative sign in E_x indicates that the field points in the –x direction, or 180° counterclockwise from the +x axis.

Q4 10 points

$$V=\frac{Q}{4\pi\varepsilon_0 R}+(\frac{-4Q}{8\pi\varepsilon_0 R})+\frac{9Q}{12\pi\varepsilon_0 R}=\frac{Q}{2\pi\varepsilon_0 R}=3.60\times10^{8}\,V$$

Q5 12 points

a)

$$Q_1=C_1V=\frac{\kappa\varepsilon_0 A}{d}V=\frac{(3.00)\times(8.85\times10^{-12}\,\mathrm{C^2/Nm^2})\times(5.00\times10^{-3}\,\mathrm{m^2})}{2.00\mathrm{mm}}(10.0\mathrm{V})=6.64\times10^{-10}\,\mathrm{C}$$

$$Q_2=C_2V=\frac{\varepsilon_0 A}{d}V=\frac{(8.85\times10^{-12}\,\mathrm{C^2/Nm^2})\times(5.00\times10^{-3}\,\mathrm{m^2})}{2.00\mathrm{mm}}(10.0\mathrm{V})=2.21\times10^{-10}\,\mathrm{C}$$

b)

$$E=\frac{V}{d}=\frac{10.0\mathrm{V}}{2.00\mathrm{mm}}=5000\mathrm{V/m\ or\ N/C}$$

$$\text{or } E=\frac{Q_1}{A\varepsilon_0}=5000\mathrm{V/m\ or\ N/C}$$

c)

$$q'=Q_2-Q_1=-4.43\times10^{-10}\,C$$

$$or\ q'=-(1-\frac{1}{\kappa})Q_2=-4.43\times10^{-10}\,C$$

Q6 10 points

For left wire: two semi-infinite $B_1=2\times\frac{1}{2}\frac{\mu_0 i}{2\pi R}=\frac{\mu_0 i}{2\pi R}$, arc $B_2=\frac{\mu_0 i\frac{\pi}{2}}{4\pi R}=\frac{\mu_0 i}{8R}$

For right wire: two semi-infinite $B_3=0$, arc $B_4=-\frac{\mu_0 i\frac{\pi}{2}}{4\pi R}=-\frac{\mu_0 i}{8R}$

So $B_{net}=\frac{\mu_0 i}{2\pi R}+\frac{\mu_0 i}{8R}-\frac{\mu_0 i}{8R}=\frac{\mu_0 i}{2\pi R}$

Q7 12 points

a) $E=0$

b) Use Gauss' law

$$E \cdot 4\pi r^2 = \frac{q_1 + \int_b^r \frac{A}{r} \cdot 4\pi r^2 \, dr}{\varepsilon_0}$$

$$E = \frac{q_1}{4\pi r^2} + \frac{A}{2} - \frac{Ab^2}{2r^2}$$

$$\Rightarrow A = \frac{q_1}{2\pi b^2} = 4.48 \times 10^{-12}\,\mathrm{C/m^2}$$

Q8 **10 points**

a) The rotating cylinder is just like a solenoid，so we can calculate the magnetic field inside it based on $B = \mu_0 n i$.

For the cylinder , the amount of charge that passes a given segment of length l on the cylinder in a time dt is $dq = \sigma l(\omega R)dt$. So the current per unit length is

$$J_S = \frac{1}{l}\frac{dq}{dt} = \sigma\omega R$$

So the field inside the cylinder is $B = \mu_0 \sigma\omega R$

b)

Consider a circular strip with width dx, a slant-distance x from the tip. The velocity of any point in this strip is $v = \omega x \sin\theta$, the current in the strip is

$$i = \frac{dq}{dt} = \frac{\sigma dx(vdt)}{dt} = \sigma\omega x \sin\theta dx$$

The magnetic field at the tip due to the strip is

$$dB = \frac{\mu_0 i (2\pi x \sin\theta)}{4\pi x^2}\sin\theta = \frac{1}{2}\mu_0 \sigma\omega \sin^3\theta dx$$

So the net magnetic field at the tip due to the cone is $\int_0^L dB = \frac{1}{2}\mu_0\sigma\omega\sin^3\theta L$